

ENVIRONMENT DIAGRAMS AND LAMBDA EXPRESSIONS

COMPUTER SCIENCE MENTORS 61A

September 16 – September 19, 2025

1 Environment Diagrams

1. Draw the environment diagram that results from running the code. What does the program print?

```
def foo(x):  
    def bar():  
        print(x)  
    return bar
```

```
x = foo(3)  
y = foo(4)  
x()  
y()
```

2. Draw the environment diagram that results from running the following code.

```
a = 1
c = 2
def b(b):
    def d():
        return b + c
    return d()
c = b(a)
a = b(c)
```

3. Give the environment diagram and console output that result from running the following code.

```
def swap(x, y):
    x, y = y, x
    return print("Swapped!", x, y)

x, y = 60, 1
a = swap(x, y)
swap(a, y)
```

2 Lambda Expressions and Higher Order Functions

1. Express the following expressions using lambdas instead of their named counterparts.

(a) `square(4)`

(b) `sum_of_squares(3, 4)`

(c)

```
def hello_world():  
    return 'Hello, world!'  
hello_world()
```

2. Make a lambda function, `make_interval()`, that takes in the upper and lower bound of an interval, and returns a function that takes in a value `x` and checks whether `x` is in the interval `[lower, upper]`, inclusive.

```
>>> make_interval = _____  
>>> in_interval = make_interval(-1, 2)  
>>> in_interval(0)  
True  
>>> in_interval(61)  
False
```

3. Draw the environment diagram that results from running the code.

```
def a(y):  
    d = 1  
    b = lambda x: y(x)  
    e = lambda x: x(3)  
    return e(b)  
d = 5  
a(lambda x: 4 - x + d)
```